Dhiraj M. Patil

Machine Learning Engineer — LLM Systems — AI Infrastructure

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Summary

Machine Learning Engineer specializing in LLM systems, including training transformer models from scratch, building data pipelines for LLM pretraining, and developing production-grade GenAI applications. Experienced with JAX/Flax, PyTorch, and the Hugging Face ecosystem for scalable model development and deployment.

Technical Skills

Languages:	Python, Java, SQL
Machine Learning:	JAX, Flax, Optax, Orbax, PyTorch, TensorFlow, Hugging Face Transformers, Hugging Face Datasets, LangChain
LLM Systems:	Transformer Architectures, LLM Pretraining, Fine-tuning, Tokenization, Retrieval Augmented Generation (RAG), FAISS, Sentence Transformers
Backend:	FastAPI, Flask, REST APIs, SQLAlchemy
Cloud:	GCP, Docker, CI/CD, GitHub Actions, Vercel
Tools:	Git, Linux, Dataset Pipelines, Model Optimization, Distributed Training

Projects

LaughLM — Mar 2026 – Present

JAX, Flax, Transformer Architecture — GitHub

- Training a decoder-only transformer LLM from scratch with modular and extensible architecture design.
- Implemented modern training techniques including RoPE positional embeddings and RMSNorm for stable training.
- Built scalable tokenization, batching, and preprocessing pipelines for large-scale text datasets for LLM pretraining.

Text Curation Pipeline — Jan 2026 – Present

Python, Hugging Face Datasets — GitHub

- Built data curation pipelines for LLM pretraining and fine-tuning workflows.
- Improved preprocessing throughput by 3× using multiprocessing-based batch transformations.
- Designed reusable and scalable dataset transformation modules.

AI Chat Platform — Oct 2025 – Dec 2025

FastAPI, Hugging Face Transformers, LangChain, GCP — GitHub

- Built a production-grade LLM platform supporting conversational AI, document Q&A, and web-assisted responses.
- Implemented Retrieval-Augmented Generation (RAG) pipelines using FAISS and Sentence Transformers.
- Deployed GPU-backed inference on GCP with a Next.js frontend on Vercel, achieving 1–3s response latency.

Open Source Contributions

Hugging Face Ecosystem

- Initiated a design discussion on reproducibility in Daggr workflows involving external model and Space dependencies across different execution environments.
- Proposed dependency provenance tracking using commit hashes and revisions to improve reproducibility, traceability, and auditability of workflow executions.
- Contribution directly led to implementation of dependency tracking features by maintainers, including merged pull requests (PRs #65, #68).

Education

University of Mumbai — Mumbai, India

Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence and Machine Learning) — 2025

CGPA: 8.38 / 10

Certifications

- **AWS Data Engineering** (AWS Certified)
- **Generative AI with AWS** (Coursera)